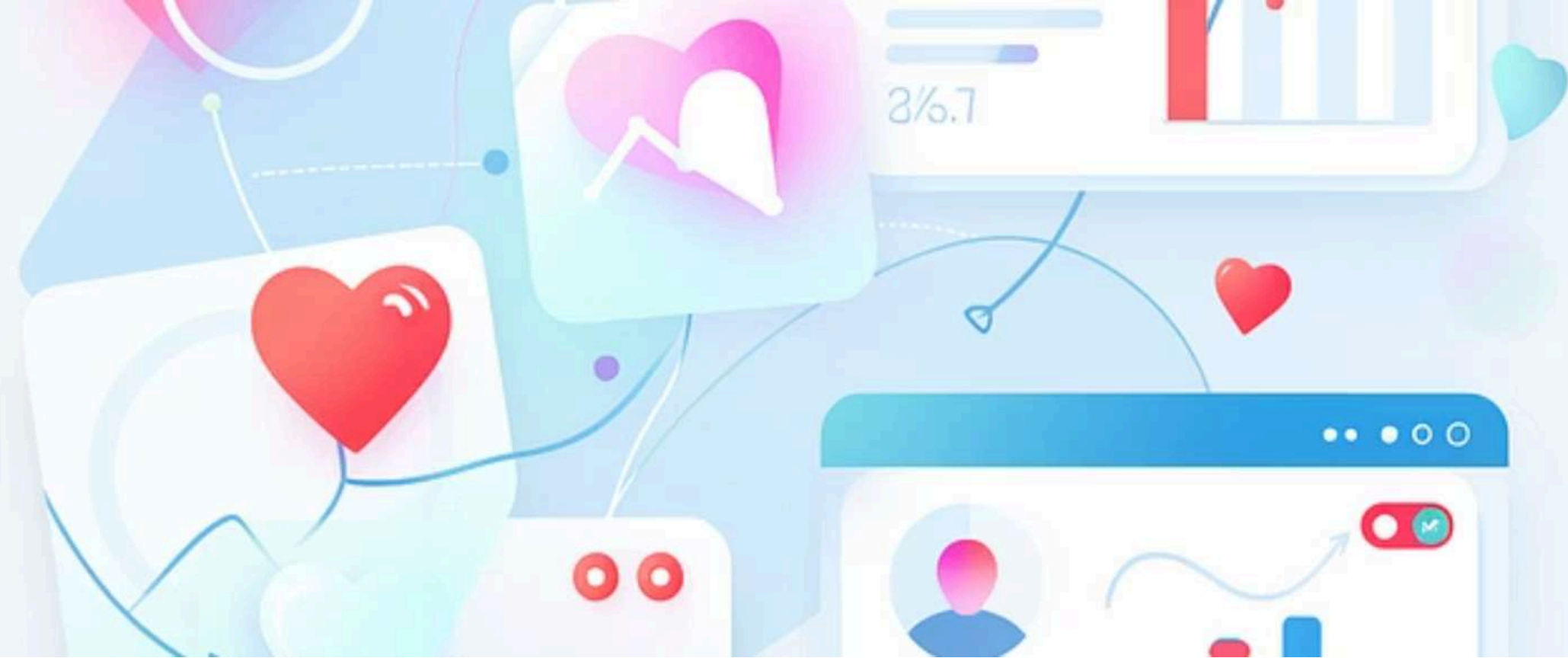




TrueLove AI

Predicting Serious Relationship Intent in Dating App Users

50K
Users Analyzed

33.2%
Serious Daters

90.6%
Model Accuracy

The Dating Dilemma: Serious vs. Casual



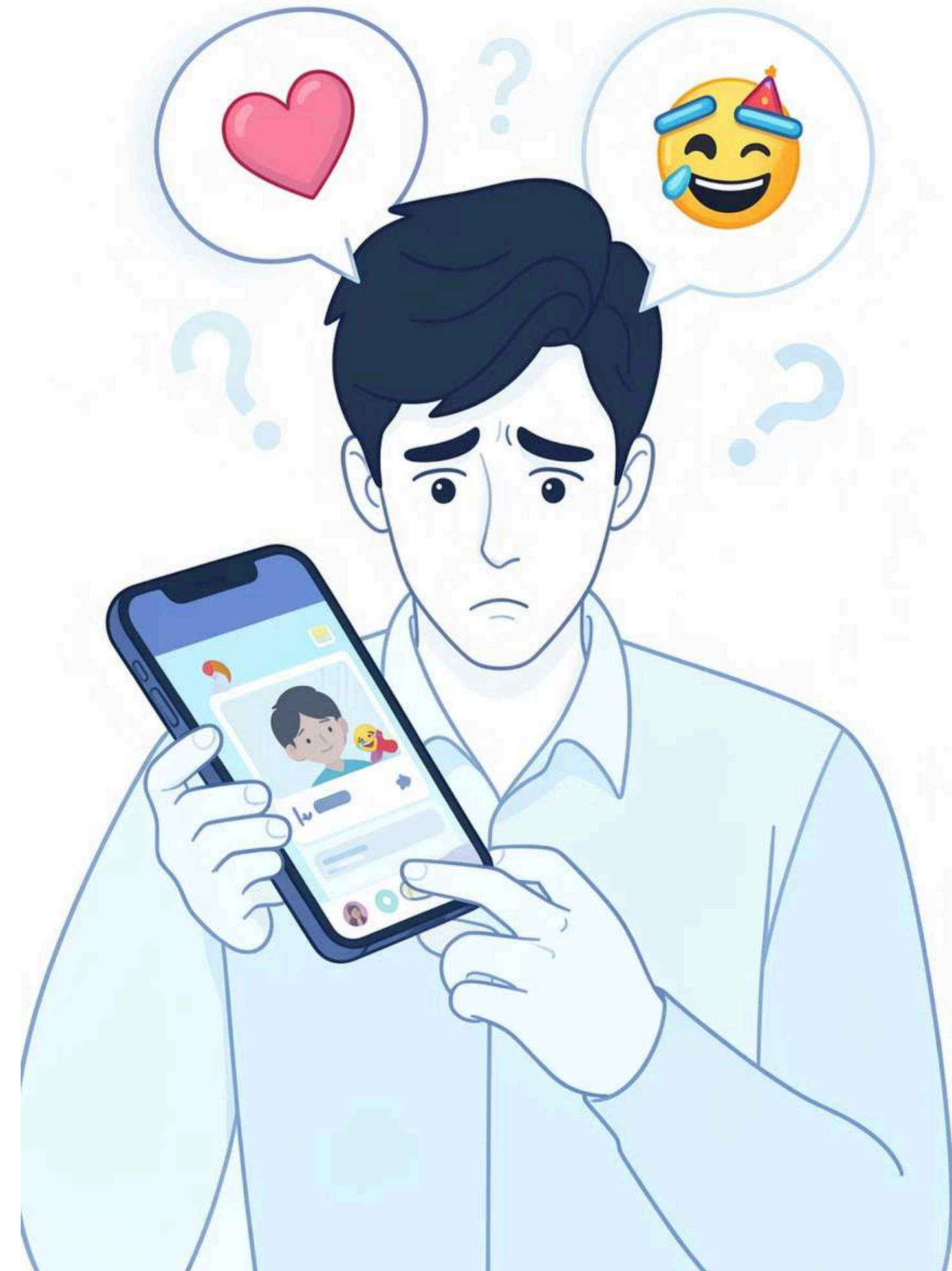
Millions of connected users, swipes, endless chats, ghosting and mixed signals.



How do we separate casual swipers from serious daters when everyone behaves differently?



Understanding user intent within raw, unstructured app behaviour to improve matching algorithms.



Introducing the Smart Dashboard

 Tying the Data Knot

Love, Life & Likes

Navigate

 Overview

 Live Predictor

 Model Comparison

 LGBTQ+ Analysis

 About

Tying the Data Knot

Predicting serious relationship intent across LGBTQ+ communities

50,000

TOTAL USERS

33.2%

SERIOUS DATERS

<https://wia1006-group8-dashboard.streamlit.app/>

Live Predictor

Enter a user's profile — Random Forest predicts their relationship intent in real time

Profile Details

Sexual Orientation

Asexual

Bio Length (characters)

248

Profile Pictures

3

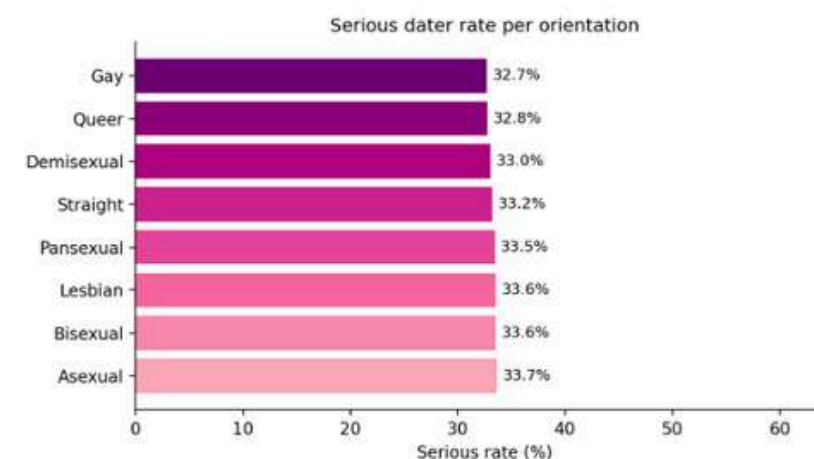
Prediction Result

 Fill in the sliders and click **Predict** to see the result.

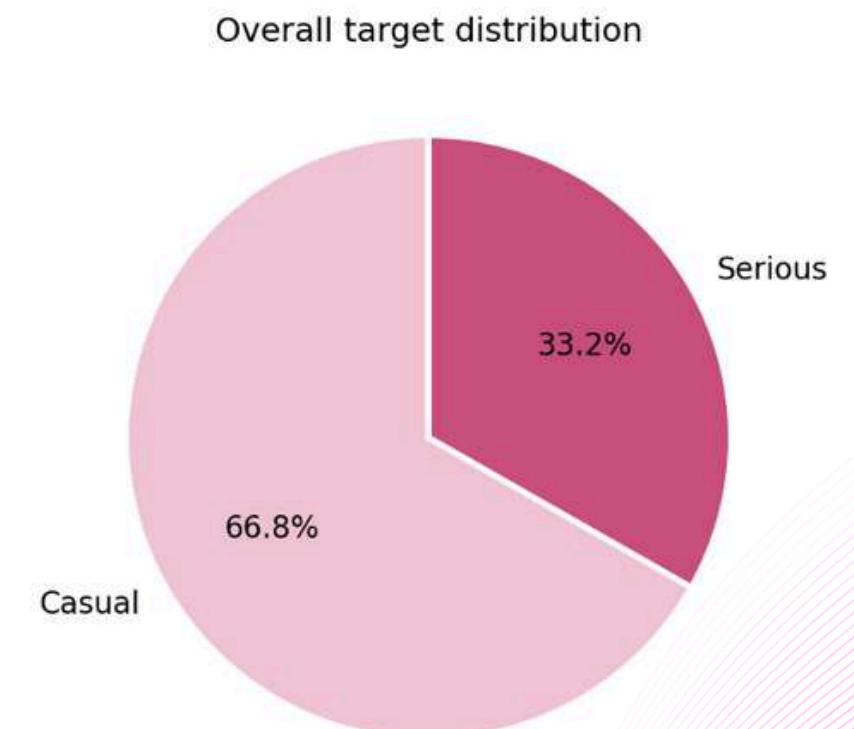
How it works: The Random Forest model was trained on 50,000 dating app users. It uses 8 behavioural features and compares each user against their own LGBTQ+ orientation group averages — not a one-size-fits-all threshold.

The score breakdown shows exactly which of the 6 Goldilocks rules the user meets or fails.

Serious rate by LGBTQ+ group



Casual vs serious split



Data Collection: Unveiling User Behavior



USER DEMOGRAPHICS

Includes Age, Gender, Location,
and 8 LGBTQ+ Orientations.



PROFILE FEATURES

Data on Bio Length, Photos
Uploaded, and Keywords.



APP INTERACTIONS

Logs of Swipes (Right/Left)
and Resulting Matches.



ENGAGEMENT METRICS

Messages Sent, Emoji Usage,
and Total App Time.

By tracking these varied behavioral and profile attributes, we gain a comprehensive picture of how users engage with dating apps, moving beyond stated preferences to reveal underlying patterns of intent.



Data Cleanup: Preprocessing for Precision

Missing Values

Remove NaNs
with rules

Feature Engineering

Create
Interest Count

Encode

Map Sexual
Orientation

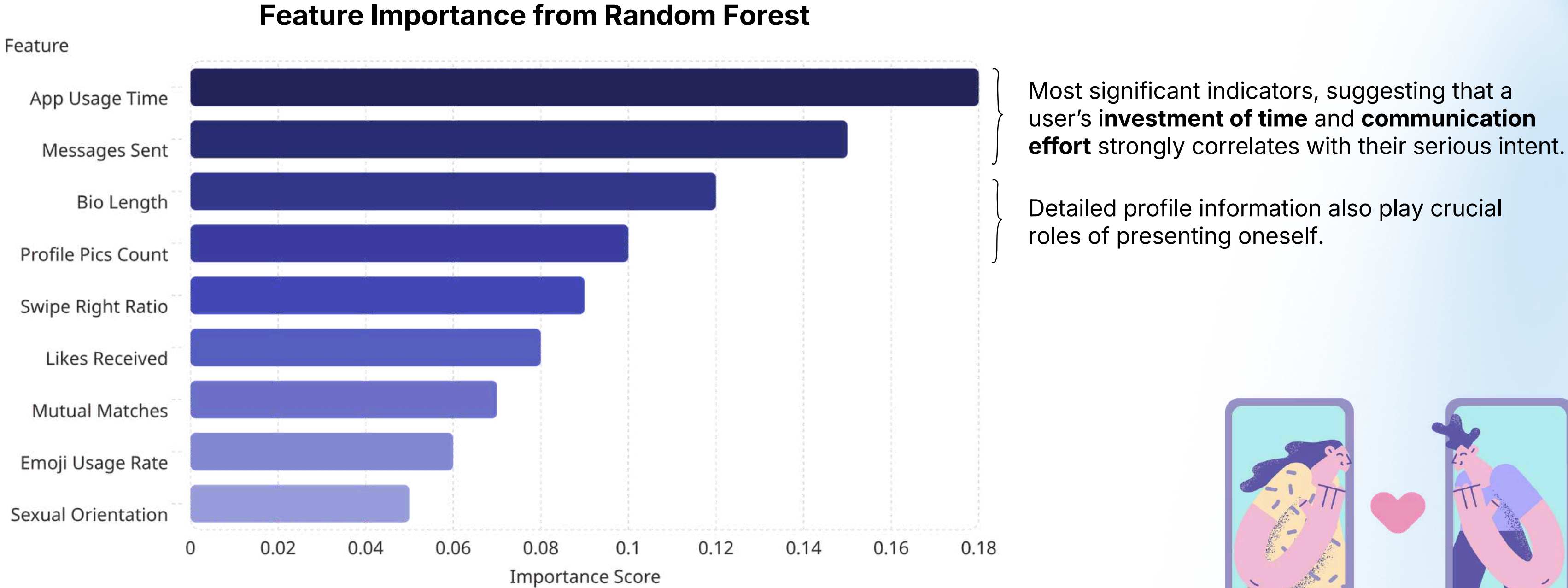
Standardize

Scale features
to unity



Result: Zero NaNs Remaining

Feature Selection: Unlocking Intent Signals



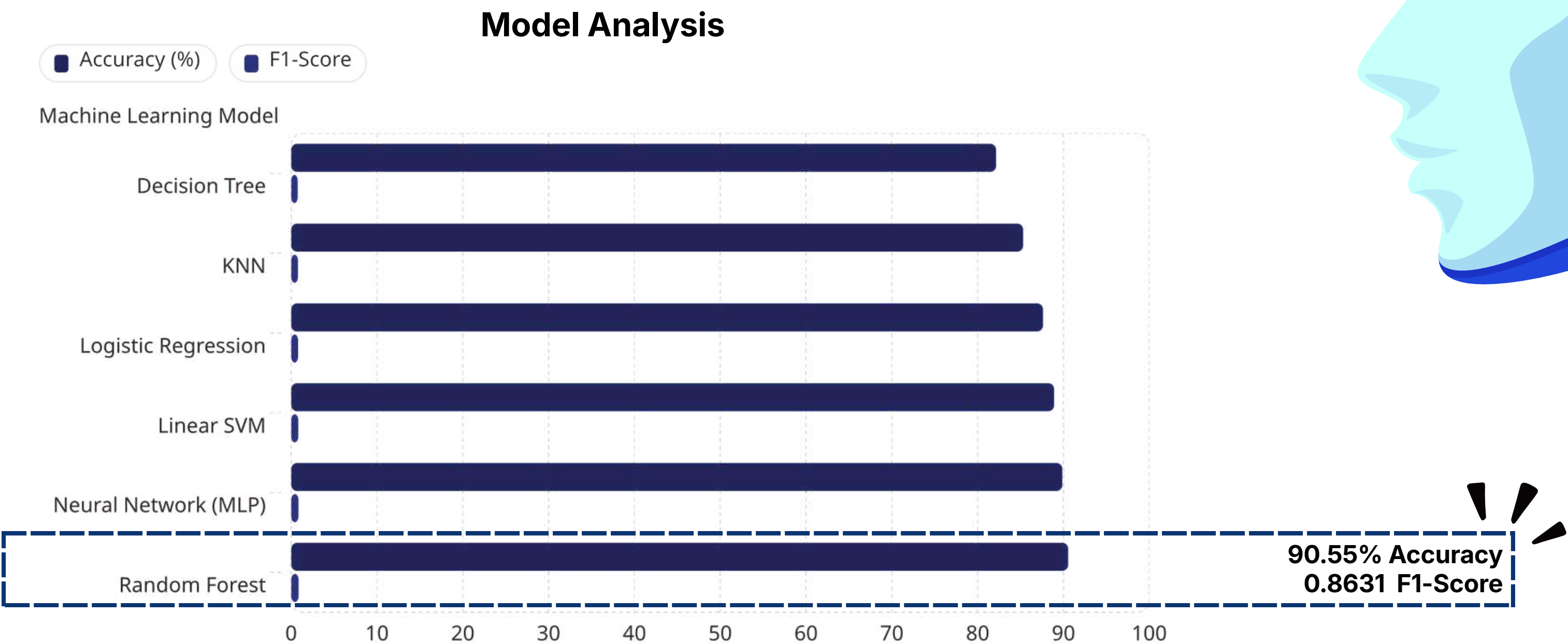


Defining User Seriousness by these Factors

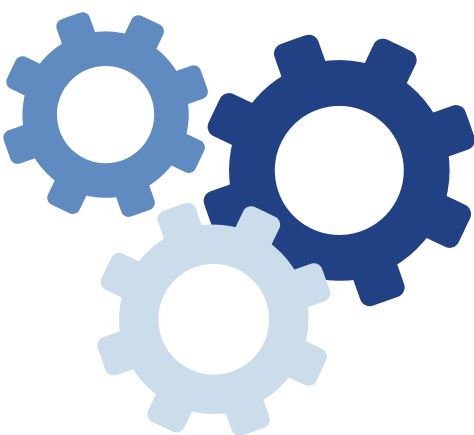
Scoring System using The Goldilocks Rule

- 1 Bio Length (Min 50 chars)**
Users investing in detailed self-description.
- 2 Swipe Selectivity (Ratio < 0.3)**
Thoughtful swiping, not just swiping right on everyone.
- 3 Message Count (Min 10/day)**
Proactive and consistent communication engagement.
- 4 Emoji Usage (Max 30%)**
Balancing expressiveness with genuine conversation.
- 5 Message Density (>0.5 msg/min)**
Sustained, active conversations per minute of app use.
- 6 Match ROI (>5 interactions/match)**
Converting matches into meaningful exchanges.
- 7 Inactivity/Resolution**
Significant periods of inactivity, being blocked, or having a formed relationship.

Model Selection & Training: Benchmarking Performance



Model Evaluation: Precision & Performance



Random Forest Model

Accuracy: 90.55%

The overall percentage of correct predictions the model made.

F1-Score: 0.8631

A balanced metric that considers both precision and recall, especially useful when classes are imbalanced.

		Actual Intent	
		Actual: Casual	Actual: Serious
Predicted Intent	Predicted: Casual	30.3k	1.66k
	Predicted: Serious	1.67k	16.4k

Precision (Casual): 0.95

Out of all users predicted as casual, 95% were truly casual.

Recall (Casual): 0.91

Out of all truly casual users, 91% were correctly identified.

Precision (Serious): 0.83

Out of all users predicted as serious, 83% were truly serious.

Recall (Serious): 0.90

Out of all truly serious users, 90% were correctly identified.



Live Predictor: Real-Time Intent Detection

New User Profile

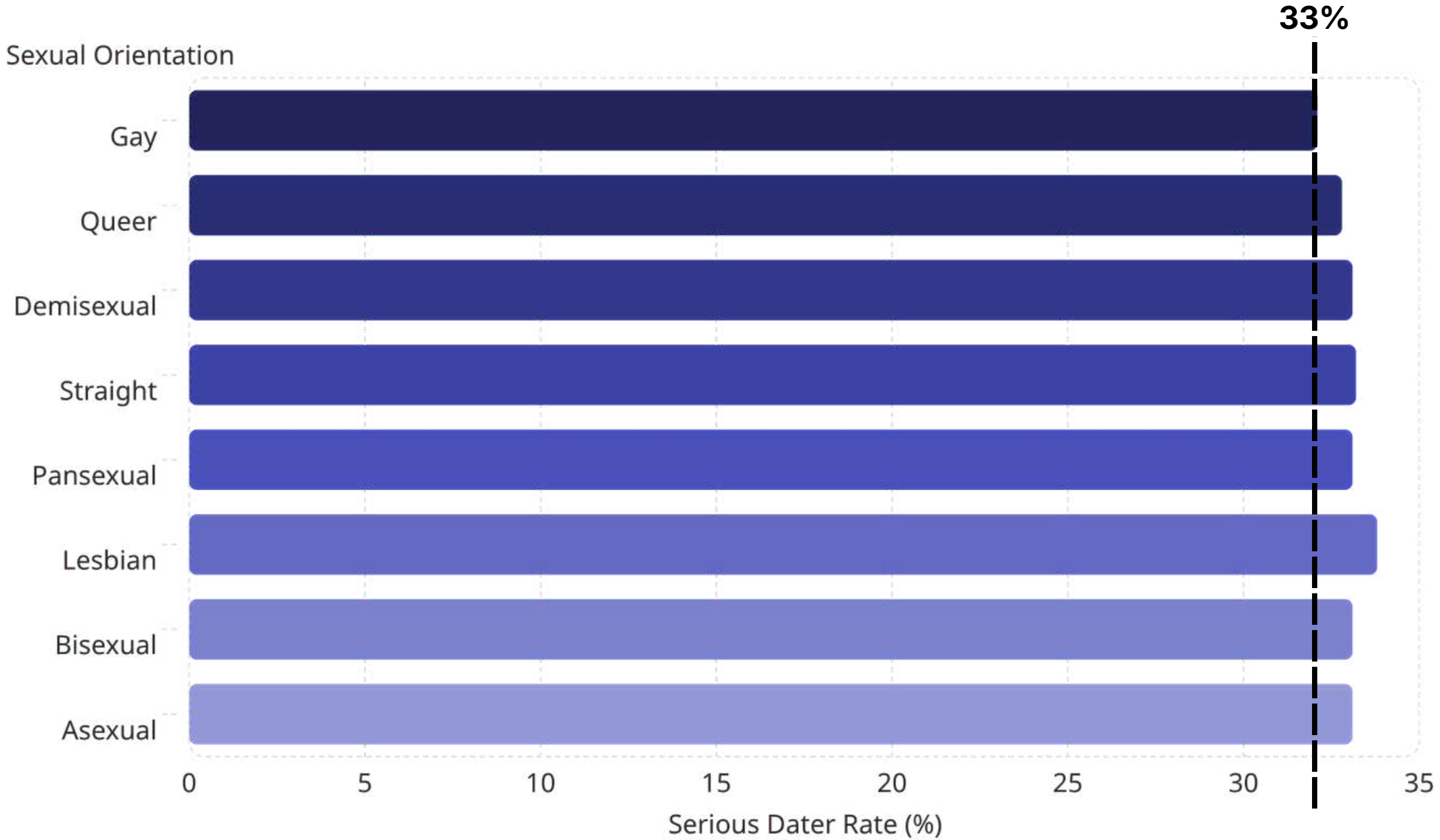
Real-Time Ingestion

RF Predicts Intent

Optimized Matching

By instantly categorizing users as serious or casual, we empower the system to tailor matching algorithms, fostering more meaningful and relevant connections.

LGBTQ+ Analysis: Intent Across Communities



The propensity for serious relationship intent is **remarkably consistent** across diverse sexual orientations.

Serious dater rates hover consistently around **33%** for all listed communities.

Highlighting a **fundamental human need** for **connection and belonging**.



Key Takeaways:

Reshaping the Future of Dating

Our data-driven approach to understanding relationship intent provides foundational insights for a more effective and inclusive dating experience.



Precision in Prediction

A 90.55% accurate model enables smarter matching and reduces user frustration.



Inclusive Connection

Serious intent is universal across all LGBTQ+ communities, transcending identity.



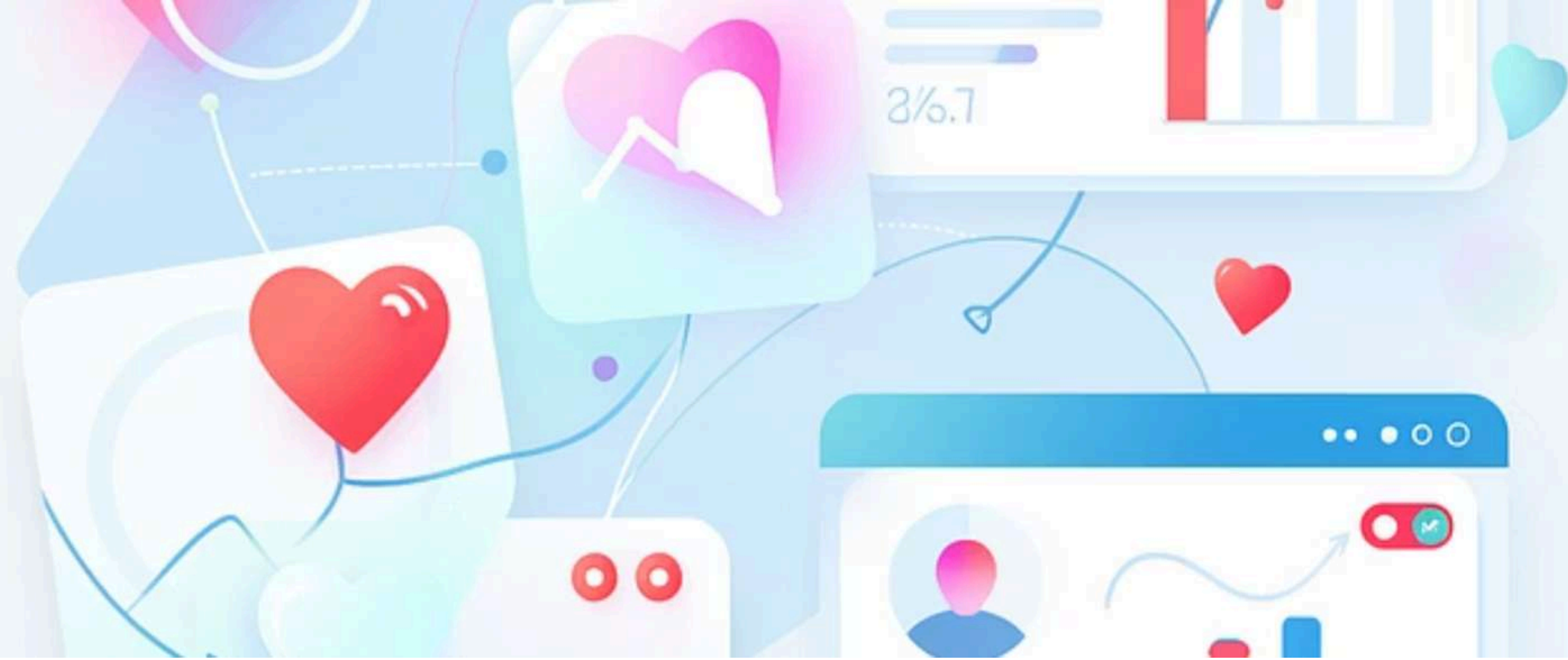
Nuanced Seriousness

The "Goldilocks Rule" defines intent through behavior, fostering genuine connections.



Dynamic Matching

Our live predictor optimizes real-time matches based on evolving user intent.



Thank You for Watching

Presented by Group 8

LEE WUEY YONG

YEW LIENN

MIAO XINYU

LIM XUAN NING

KILANI WASIM

MOHAMMED NASIR ELSEED MUJAHID

Links to Our Project

Dating App Behavior Dataset 2025

<https://www.kaggle.com/datasets/keyushnisar/dating-app-behavior-dataset/code>

Colab Notebook for Codes

<https://colab.research.google.com/drive/1J78eyD5oM74rbbe8Kcu9ITqCIngLk8j2>

Our 5 Minute Presentation Video

https://drive.google.com/drive/folders/1hkwp-kC_6W-YDdAz983EyLlpFbQpU65r

Our GitHub Repository (TrueLove AI Dashboard)

https://github.com/yewlienn-2025/WIA1006_Submission